

CreditLens

Explainable credit risk — from prediction to underwriting decision.

144

ENGINEERED FEATURES

0.7285

ROC-AUC

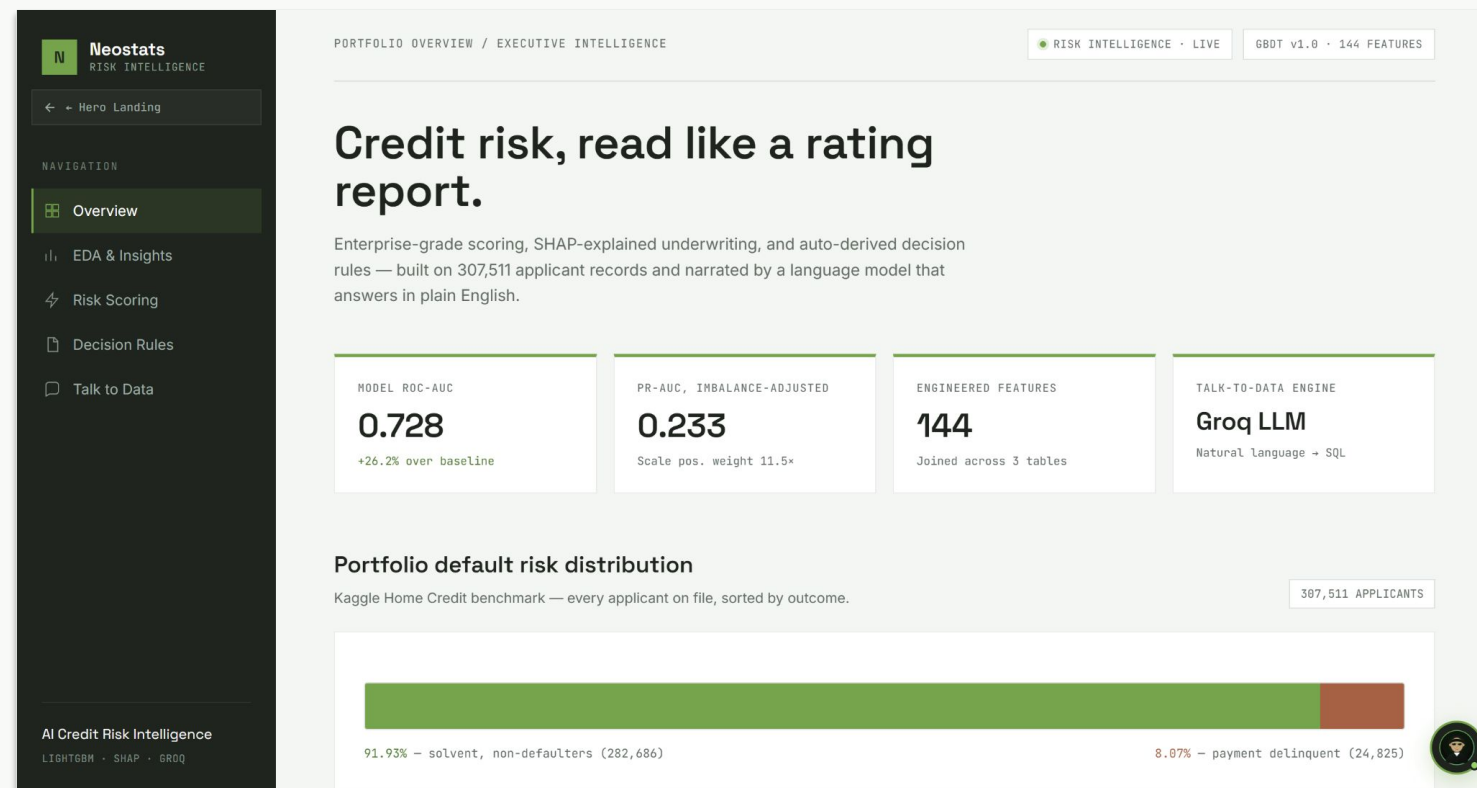
LightGBM + SHAP + Talk-to-Data

A working risk-intelligence interface built around the model.

PORTFOLIO OVERVIEW

EXPLAINABLE

ACTIONABLE



Default is rare. A wrong decision is not.

The portfolio is highly imbalanced, so model quality must be judged beyond accuracy.

8.07%

DEFAULT RATE

11.5 : 1

CLASS RATIO

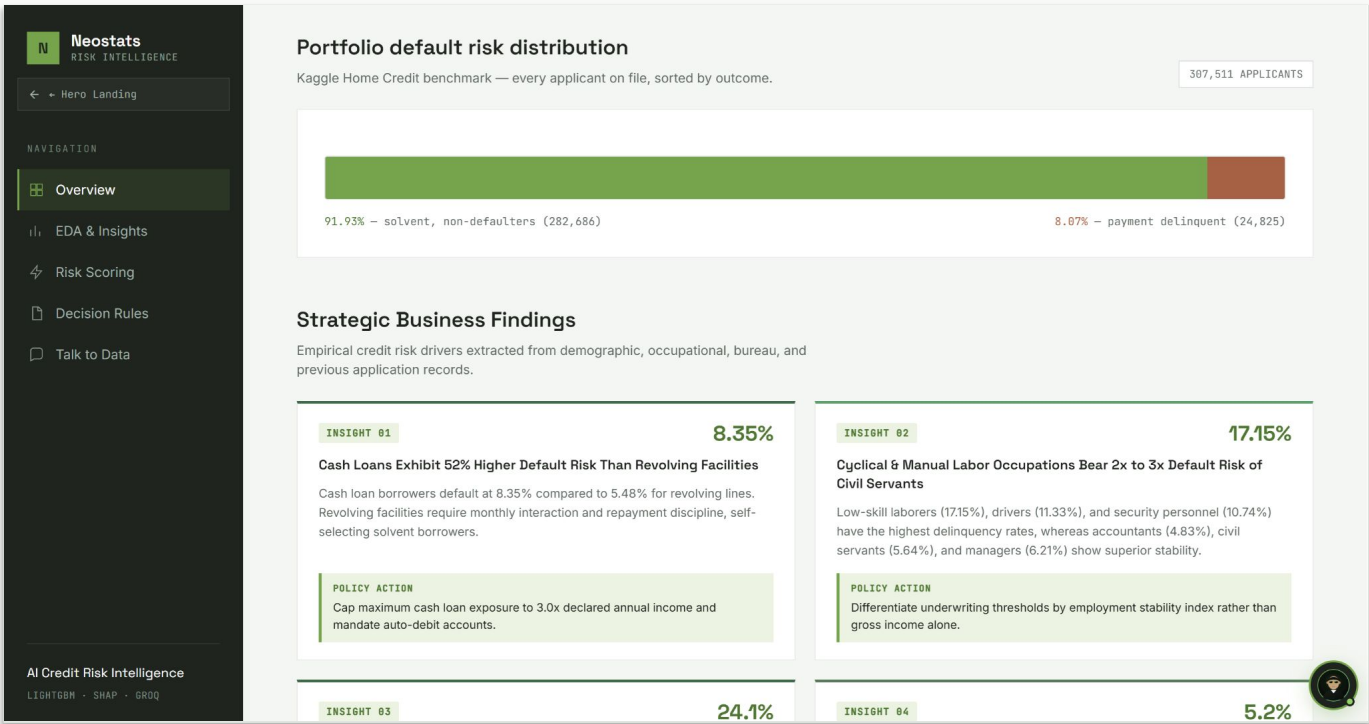
PR-AUC

PRIMARY SIGNAL

Why the imbalance matters

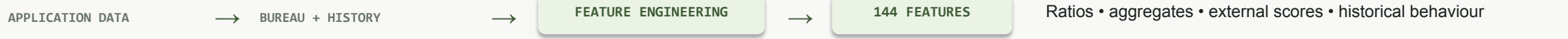
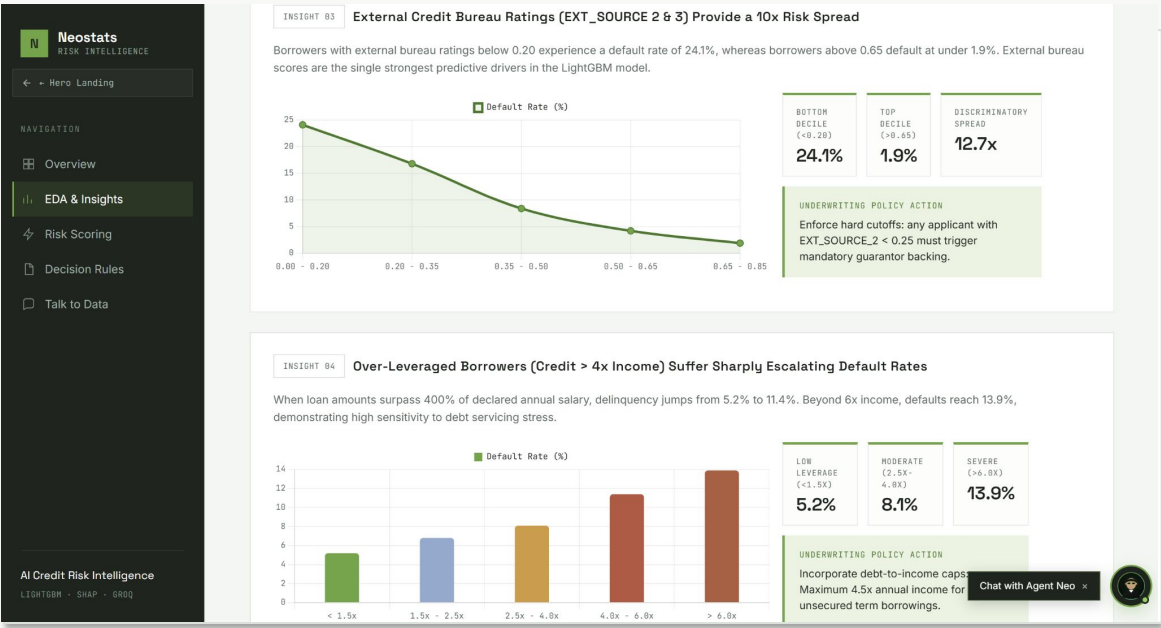
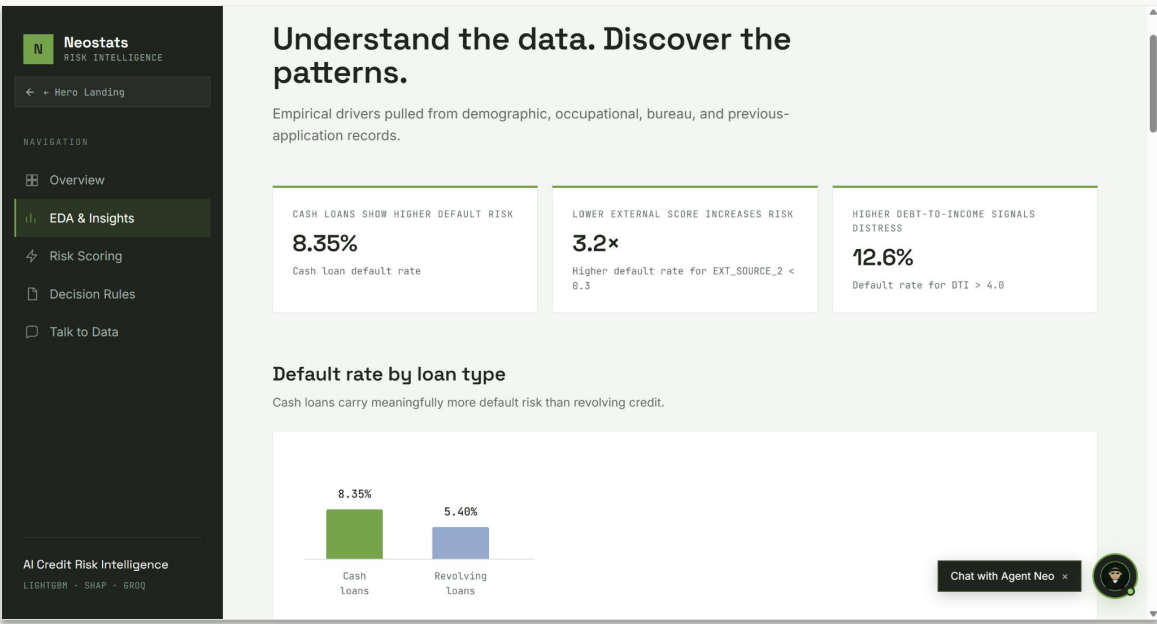
- Cost-sensitive learning**
the minority class needs more weight.
- Stratified 5-fold CV**
keeps default cases represented in validation.

DECISION-FIRST



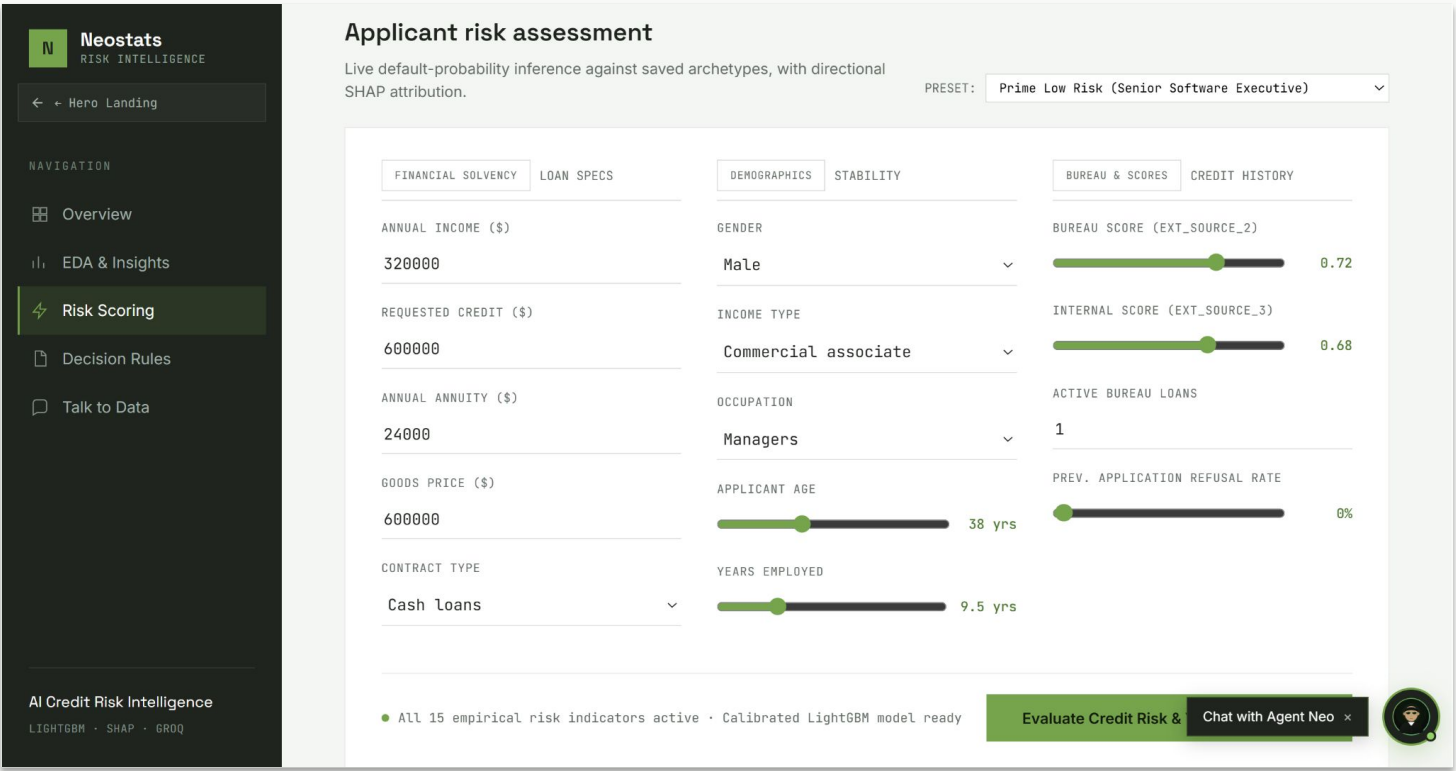
The raw portfolio becomes model-ready evidence.

EDA identifies the signals worth carrying into feature engineering and underwriting.



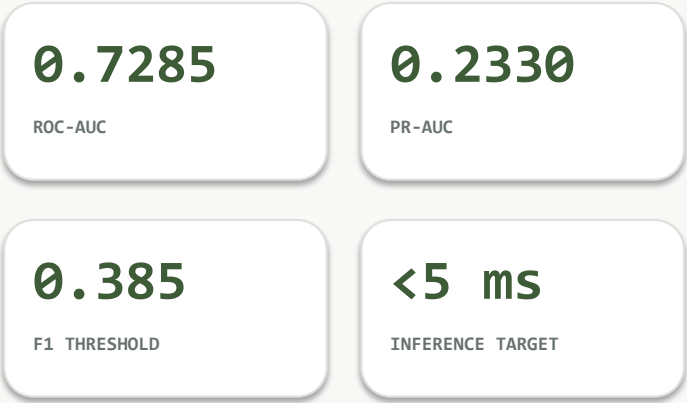
The score is only useful when it reaches a decision.

LightGBM ranks risk; the operating threshold converts probability into an underwriting action.



LIGHTGBM

800 estimators • max depth 7



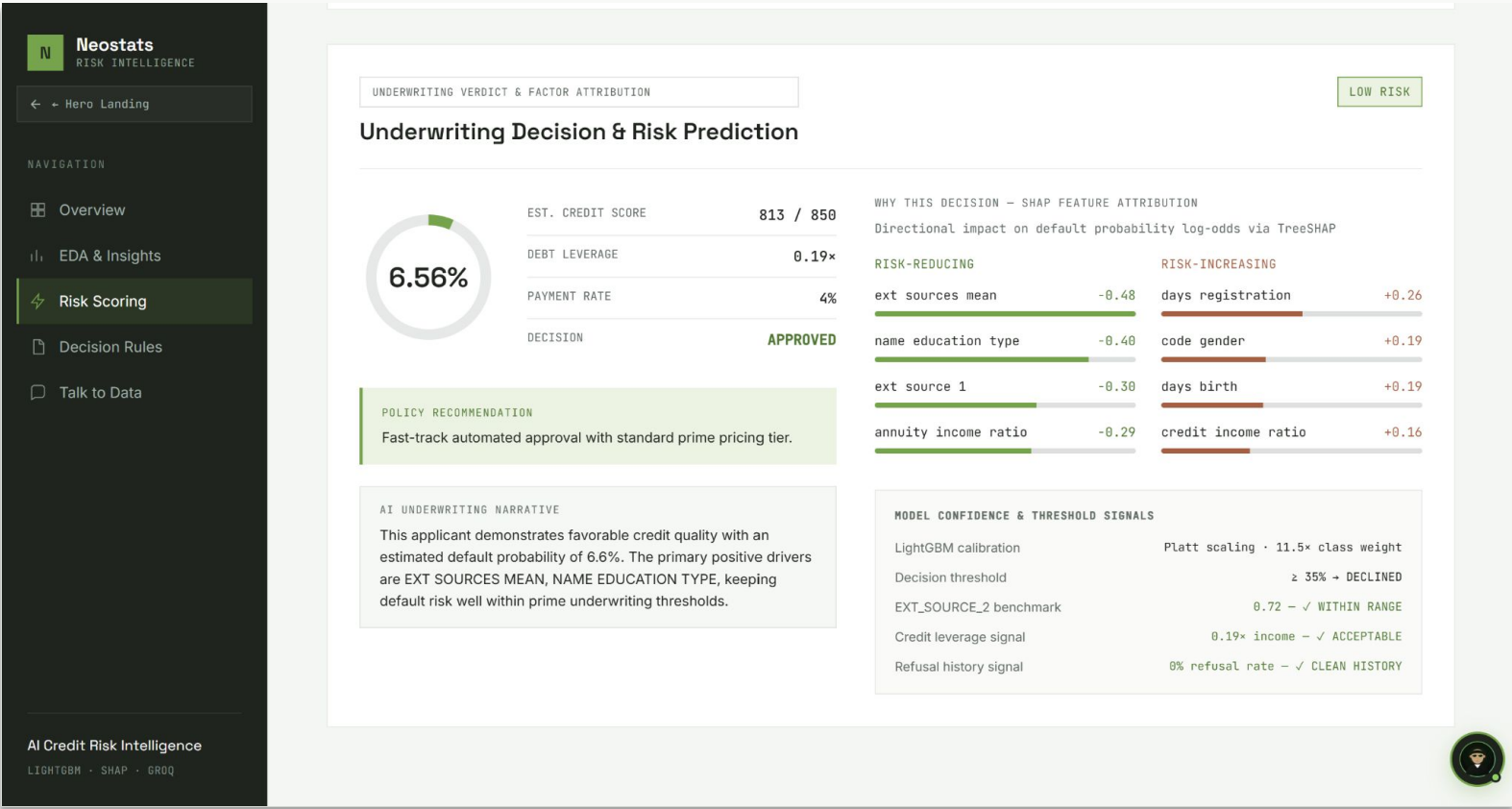
MODEL FLOW

features → score → threshold → decision

EARLY STOPPING

A prediction should be inspectable.

SHAP shows which applicant attributes pushed an individual risk score higher or lower.



WHAT MOVES RISK?

EXT_SOURCE_2 / 3

external score signals

DAYS_EMPLOYED

employment history

CREDIT_INCOME_RATIO

repayment pressure

LOCAL EXPLANATION

The model can explain one applicant — not only the portfolio average.

Risk signals become repeatable actions.

Seven policy rules translate model evidence into a consistent underwriting path.

N

Neostats

RISK INTELLIGENCE

←

Hero Landing

NAVIGATION

Overview

EDA & Insights

Risk Scoring

Decision Rules

Talk to Data

AI Credit Risk Intelligence

LIGHTGBM • SHAP • GBDT

Strategic Business Findings

Empirical credit risk drivers extracted from demographic, occupational, bureau, and previous application records.

INSIGHT 01

8.35%

Cash Loans Exhibit 52% Higher Default Risk Than Revolving Facilities

Cash loan borrowers default at 8.35% compared to 5.48% for revolving lines. Revolving facilities require monthly interaction and repayment discipline, self-selecting solvent borrowers.

POLICY ACTION

Cap maximum cash loan exposure to 3.0x declared annual income and mandate auto-debit accounts.

INSIGHT 02

17.15%

Cyclical & Manual Labor Occupations Bear 2x to 3x Default Risk of Civil Servants

Low-skill laborers (17.15%), drivers (11.33%), and security personnel (10.74%) have the highest delinquency rates, whereas accountants (4.83%), civil servants (5.64%), and managers (6.21%) show superior stability.

POLICY ACTION

Differentiate underwriting thresholds by employment stability index rather than gross income alone.

INSIGHT 03

24.1%

External Credit Bureau Ratings (EXT_SOURCE 2 & 3) Provide a 10x Risk Spread

Borrowers with external bureau ratings below 0.20 experience a default rate of 24.1%, whereas borrowers above 0.65 default at under 1.9%. External bureau scores are the single strongest predictive drivers in the LightGBM model.

POLICY ACTION

Enforce hard cutoffs: any applicant with EXT_SOURCE_2 < 0.25 must trigger mandatory guarantor backing.

INSIGHT 04

5.2%

Over-Leveraged Borrowers (Credit > 4x Income) Suffer Sharply Escalating Default Rates

When loan amounts surpass 400% of declared annual salary, delinquency jumps from 5.2% to 11.4%. Beyond 6x income, defaults reach 13.9%, demonstrating high sensitivity to debt servicing stress.

POLICY ACTION

Incorporate debt-to-income caps: Maximum 4.5x annual income for unsecured term borrowings.

INSIGHT 05

6.85%

INSIGHT 06

11.8%

DECISION TIERS

LOWstandard flow

MEDIUMreview / checks

HIGHdecline / escalate

7 RULES

Policy makes the model operational.

Ask the portfolio a question.

Natural language becomes constrained SQL, then returns a business-ready answer.

N

Neostats

RISK INTELLIGENCE

← Hero Landing

NAVIGATION

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AI Credit Risk Intelligence

LIGHTGBM • SHAP • GROQ

NEOSTATS AI ANALYST

POWERED BY GROQ LLAMA 3.3 • SQLITE LIVE

Portfolio Conversational Intelligence

Query 307,511 loan records in natural English. Generates verified SQL and provides executive takeaways.

SESSION CONVERSATION THREAD

SQLITE PORTFOLIO

Clear Chat

INTERACTIVE AUTO-CHART: DEFAULT RATE PCT

BY NAME INCOME TYPE

NAME INCOME TYPE	TOTAL APPLICANTS	DEFAULT RATE PCT
Unemployed	6	
Working	26010	9.46

Chat with Agent Neo

THE FLOW

- 01

ASK
- 02

GENERATE
- 03

VALIDATE
- 04

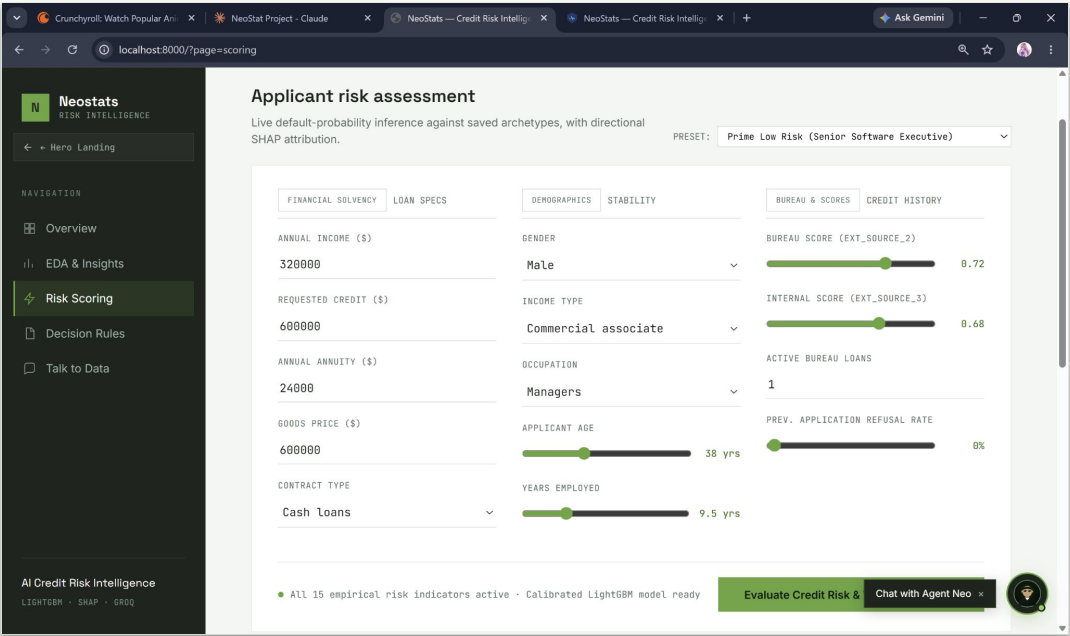
ANSWER

SELECT-only

mutation blocker • schema validator • auto-LIMIT

The finished product is more than one dashboard.

Different screens answer different questions — assessment, findings and conversational analysis.

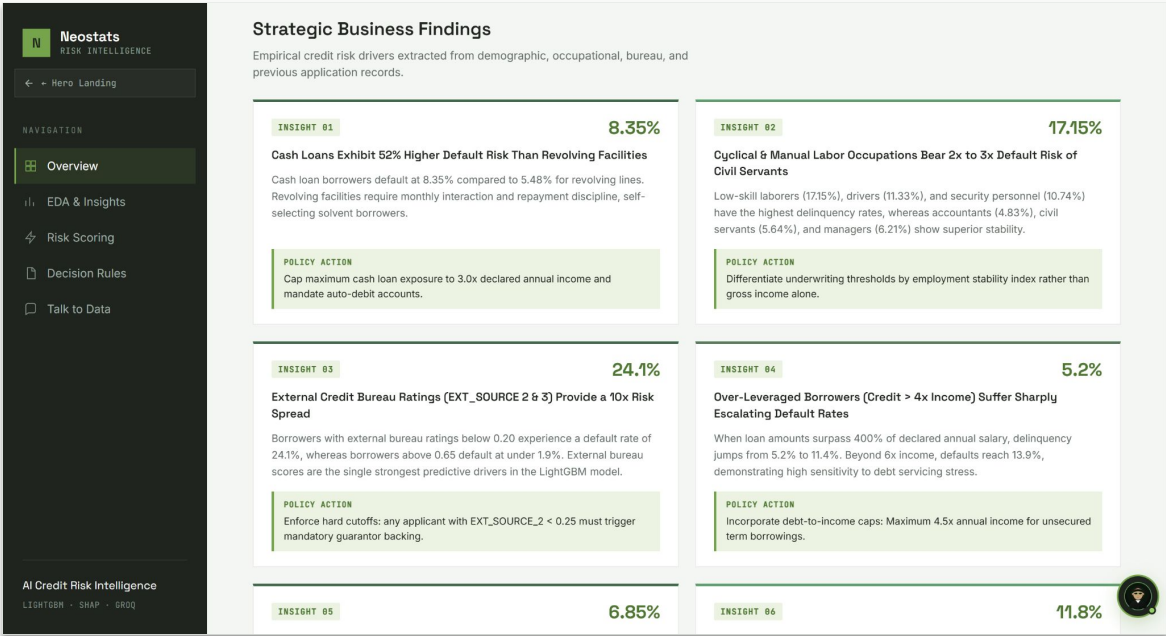


APPLICANT RISK ASSESSMENT

Decision support

STRATEGIC BUSINESS FINDINGS

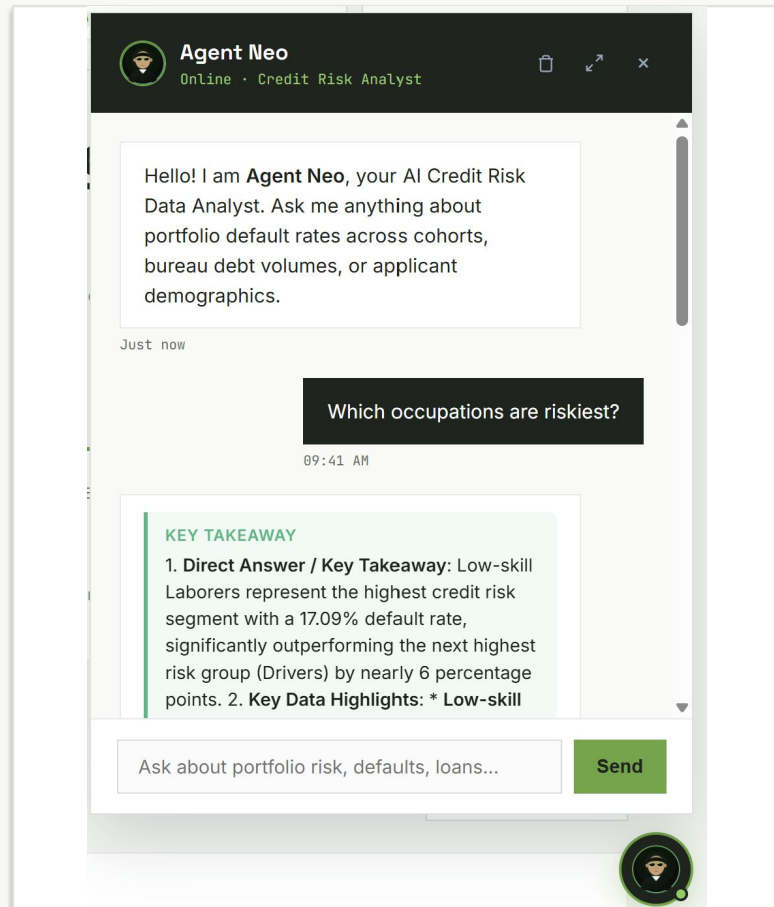
Portfolio interpretation



The product moves from an individual borrower to a portfolio-level view without leaving the same system.

A conversational layer on top of the risk system.

Agent Neo gives the user a natural-language entry point into portfolio intelligence.



WHAT THE AGENT ADDS

01

NATURAL LANGUAGE

Ask for insights without writing SQL.

02

DATA ACCESS

Connect questions to portfolio tables.

03

GUARDRAILS

Keep generated queries constrained.

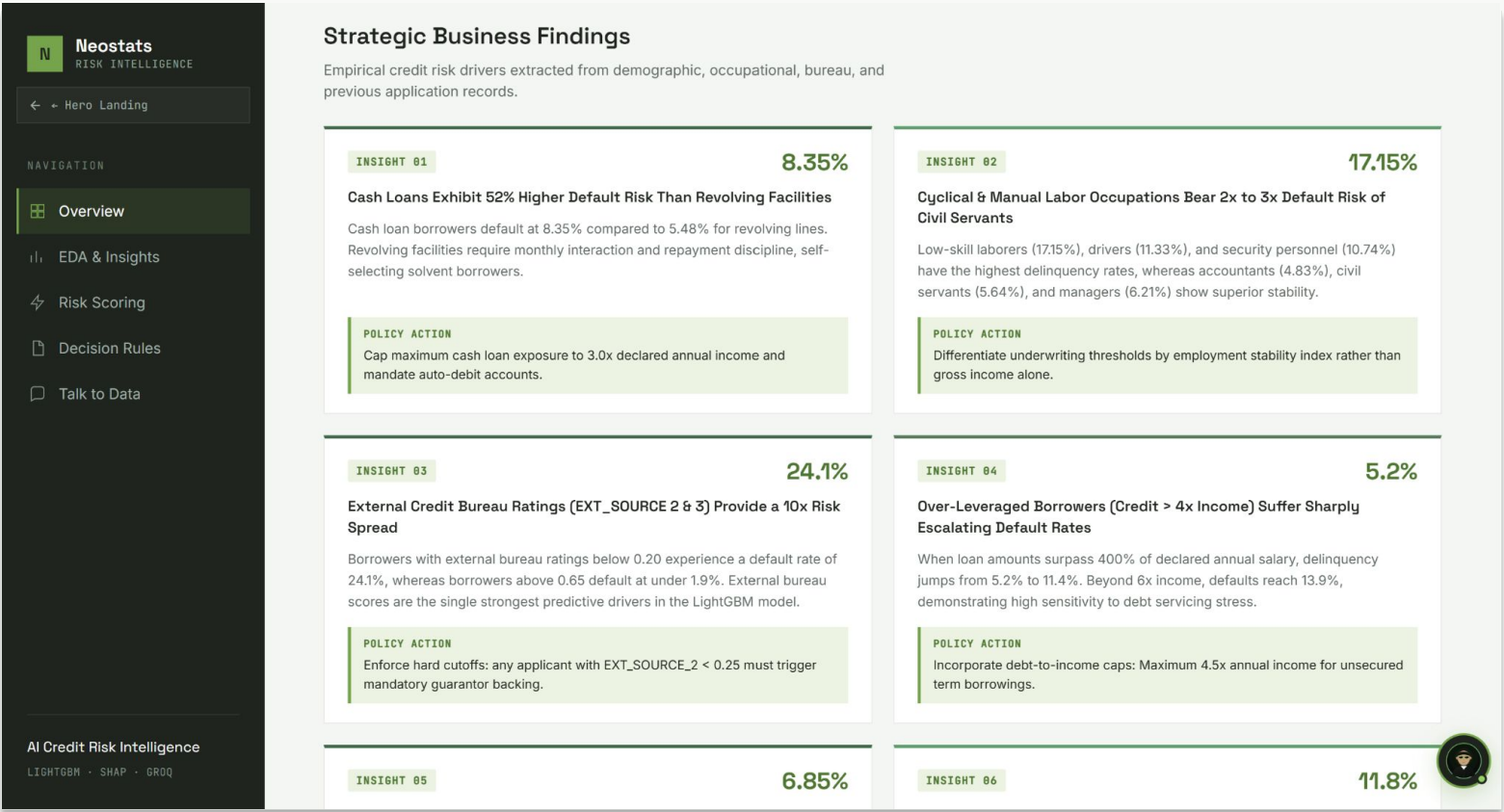
GROQ

SELECT-ONLY

AUTO-LIMIT

The system turns model output into portfolio insight.

Strategic findings make the risk model useful beyond a single applicant.



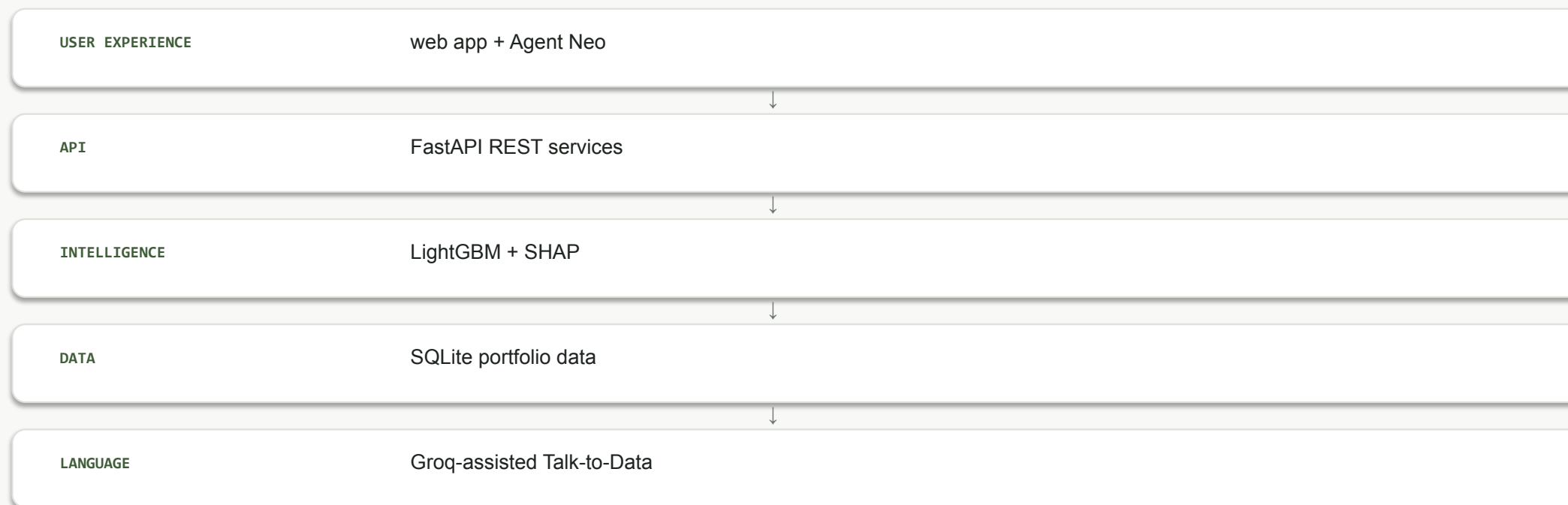
FROM MODEL TO MANAGEMENT

- Identify concentrations
see where risk is accumulating.
- Prioritize action
focus review on material segments.
- Explain the why
connect findings back to model evidence.

PORTFOLIO LEVEL

A compact stack connects the intelligence end-to-end.

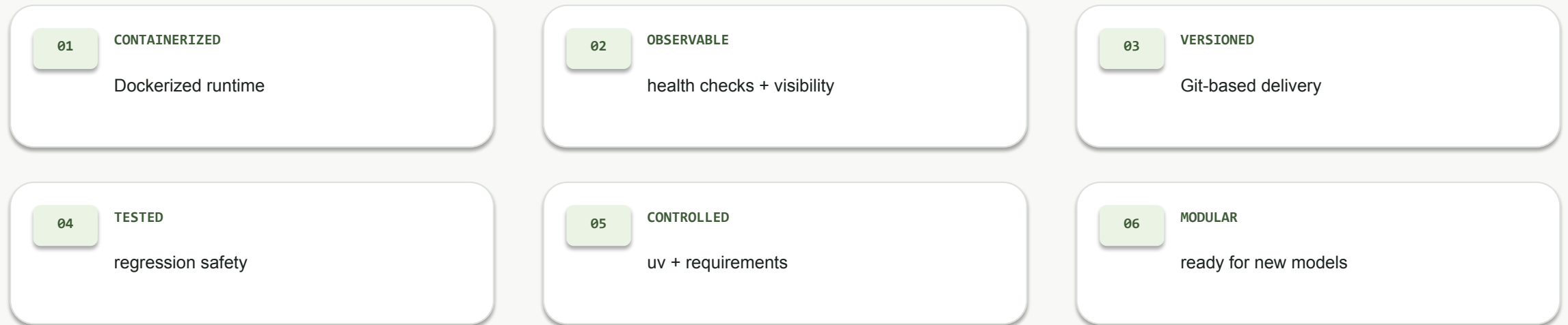
The system separates interface, API, model, data and language services.



Simple enough to understand. Modular enough to extend.

Built to run, not just demo.

Reliability comes from reproducible deployment and disciplined delivery.

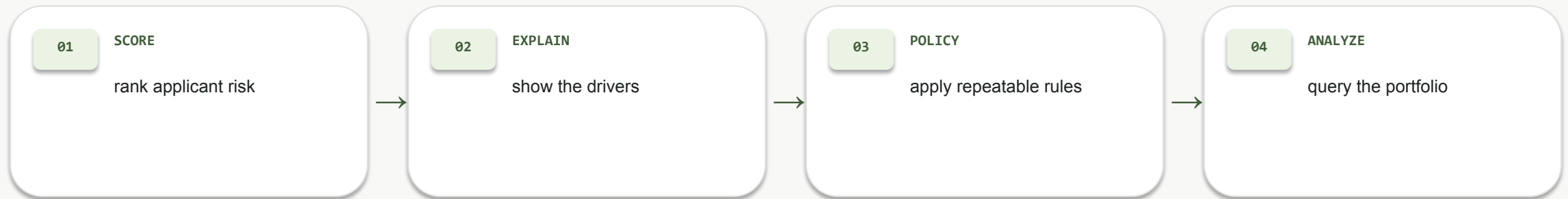


DELIVERY

reproducible • observable • testable • maintainable

One workflow replaces disconnected risk tasks.

The value is the connection between model evidence and the action that follows.



WHAT THE USER GETS

— Faster decisions

less switching between tools and views.

— More transparent decisions

model output comes with an explanation.

— Portfolio awareness

individual decisions connect to broader risk patterns.

Explainable credit risk, built for action.

Prediction is only the beginning.
The system connects score → explanation → policy → portfolio insight.

LIGHTGBM

SHAP

POLICY

TALK-TO-DATA

Next: retraining • fairness auditing • streaming intelligence

CORE WORKFLOW

SCORE



EXPLAIN



APPLY POLICY



ASK THE PORTFOLIO